Statistics

⊙ Population Data: Collection of all Items of interest to our study.

Denoted with 'N'.

Number we obtain when using a Population are called Parameters.

⊙ Sample Data: It is a Subset of Population Data.

Denoted with 'n'.

Number we obtain when working with a sample are called Statistics.

⇒ Population Data is hard to define and hard to observe in Real life.

⇒ Sample Data is much easier to gather.
Less time Consuming
Less Costly.

⇒ Sample Data has two defining Characteristics:

↳ Randomness : A Random Sample is collected when each member of the sample is chosen from the population strictly by Chance.

↳ Representiveness : A Representative Sample is a Subset of the population that accurately reflect the members of the entire population.

Descriptive Statistics

=> We can classify Data in two main Ways :

⊙ Types of Data ⊙ Measurement Level

↳ Categorical
↳ Numerical

↳ Categorical Data : Describes Categories or Groups.

e.g. Car Brands,
Yes/No questions.

↳ Numerical Data : Describes and represents Numbers.

Numerical Data is divided in
↳ Discrete : Usually can be counted in a finite manner.

e.g. No. of Children,
Exam Scores.

↳ Continuos : Continuous Data is Infinite and ~~is~~ impossible to Count.

e.g. Weight, Height, Area, Distance, Time.

i.e. Continuous Variable can vary by incomprehensibly small amounts. ~~and~~

① Measurement Level :

↳ Qualitative Data : Qualitative Data can be Nominal or Ordinal.

↳ Nominal : They are categories that can't be Ordered.

e.g. ~~Era ls~~ Car brands, Seasons., Gender.

i.e. They can't be ordered. We can't say one is better than another.

↳ Ordinal : Consists of Groups or Categories that follows a Strict order.

e.g. Food : bad,
medium,
good.

i.e. We can order these Categories:

bad < medium < good.

↳ Quantitative Data : Quantitative Data Can be Interval or Ratio.

↳ Interval : Doesn't have a true Zero.

↳ Ratio : Heve a True Zero.

i.e. Temperatures in Celcius & Fahrenheit are interals as they don't have a true Zero.

Temperature is ~~K~~ Kelvin
is Ratio as it has a true
Zero.

Visualization Techniques

⊙ Categorical Variables :

<1> Frequency Distribution Table:

Has two Columns. Category and it's Frequency.

i.e.

	Frequency
Audi	124
BMW	98
Mercedes	113
Total	335

<2> Bar Chart Or Column Chart :

Visual representation Of Frequency Distribution table.

<3> Pie chart : It shows reletive Frequency.

i.e. How many numbers Of each categories Of cars are sold as part Of Total sale.

All ~~relete~~ relative frequency in Pie chart add upto 100%.

especially useful when we want to see share of items in Total.

<4> Pareto Chart: It is a Special type of Bar chart where categories are shown in Descending order of Frequency.

It has bars in Descending Order of Frequency with a Curve on it showing Cummulative Frequency.

It is kind of a Mix ~~between~~ of Bar chart and ~~Relative~~ Pie Chart.

(·) Numerical Variables:

<1> Frequency Distribution Table:

Data is divided in Intervals and each Interval is associated with frequency of Data falling in the Interval.

<2> Histogram : Looks like a Bar chart.

It shows frequency Of different Intervals.

Unlike Bar charts, Bars in Histogram ~~are~~ touches each other as Intervals are continuous.

<3> Cross Tables and Scatter Plots :

Used to plot relationship between two Variables.

↳ Cross tables or Contingency tables:
It is used for Categorical variables.

i.e.

Investors

Type of Investment	Investor A	B	C	Total
Stocks	96	185	39	320
Bonds	181	3	29	213
Real Estate	88	152	142	382
Total	365	340	210	915

Side by Side Bar Chart is used to Visualize Cross tables.

↳ Scatter Plot :
Used for numerical variables.

One variable is plotted on X-axis and another one is plotted on Y-axis.

Generally used to examine how data is distributed.

Measures Of Central Tendency

<1> Mean :

Also known as Simple Average.

Denoted by :

For Population : μ
For Sample : $\bar{x}$

Found by adding up all the components and then dividing by the number of Components.

$$\mu \text{ or } \bar{x} = \frac{\sum_{i=1}^{N} x_i}{N}$$

Downside of Mean is, it is easily affected by Outliers.

<2> Median : It is the Middle number in Ordered Dataset.

$$\text{Median} = \text{No. at Position } \frac{n+1}{2}$$

(3) Mode: It is the Value that Occurs most Often in ~~Datset~~ Dataset.

(can be used for both Numerical and Categorical data)

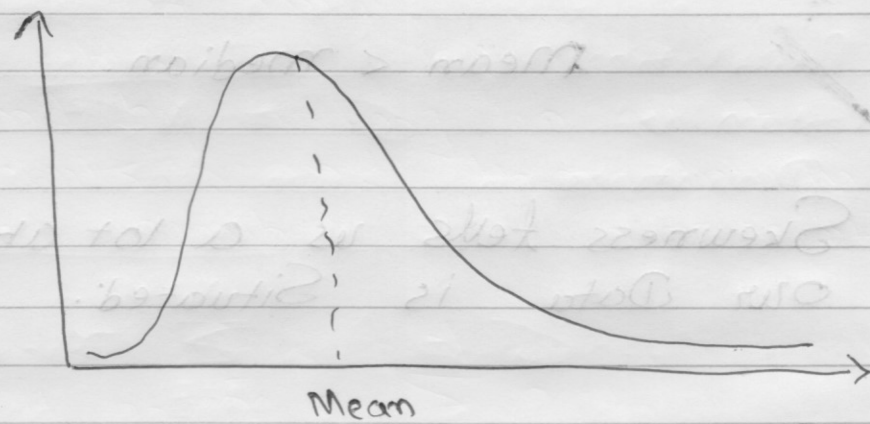
We can have more than One mode in Dataset.

-> Measures of Central ~~For~~ Tendency Should be Used together rather than Independently.

Measures of ~~Assymetry~~ Asymmetry

Skewness :

Skewness indicates whether the Observations in a dataset are concentrated on one Side.



(Right or Positive Skewness)

Right Skewness means outliers are to the right or data has long tail to the right.

Left or Negative Skewness means that outliers are to the left or data has long tail to the left.

↳ For Right or Positive Skew:

Mean > Median.

↳ For Zero Skewness:

Mean = Mode = Median

↳ For Left or Negative Skew:

Mean < median.

Skewness tells us a lot about where our Data is Situated.

Formula For Skewness:

$$\frac{\frac{1}{n}\sum_{i=1}^{n}(x_i-\bar{x})^3}{\sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i-\bar{x})^2}^{\,3}}$$

Measures of Variability

<I> Variance : Variance measures the dispersion of data points arond their Mean.

Population Variance $\sigma^2 = \dfrac{\sum_{i=1}^{N} (x_i - \mu)^2}{N}$

μ = Population Mean
N = No. of observations in Population.

Sample Variance $s^2 = \dfrac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n-1}$

$\bar{x}$ = Sample mean
n = no. of Observations in Sample.

<2> Standard Deviation :

It is the Square Root of Variance.

Used more frequently as value of Variance tends to be large due to squares.

Population Standard Deviation $\sigma = \sqrt{\sigma^2}$

$$\therefore \sigma = \sqrt{\frac{\sum_{i=1}^{N} (x_i - \mu)^2}{N}}$$

Sample Standard Deviation $S = \sqrt{S^2}$

$$\therefore S = \sqrt{\frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n-1}}$$

<3> Coefficient of Variation (CV):

It is Standard Deviation divided by Mean.

Also known as Relative Standard Deviation.

Population CV $\quad C_v = \dfrac{\sigma}{\mu}$

Sample CV $= \hat{C}_v = \dfrac{s}{\bar{x}}$

~~Coef~~

Coefficient of Variation is used frequently when comparing different Datasets as it has no unit.

Measures of Relationship between Variables

<I> Covariance :

When two variables are correlated, one of the main statistics to measure this correlation is Covariance.

Covariance is a measure of the joint variability of two variables.

Covariance can take values from $-\infty$ to $+\infty$.

- A positive Coviance means that the two variables move together.
- Coviance of 0 means that the two variables are Independent.
- A negative Coviance means that the variables move in Opposite Direction.

~~Covariance can tak~~

Covariance can take values from $-\infty$ to ∞ and sometimes, values are too small or too large to put into Perspective.

Sample Covariance $S_{xy} = \sum_{i=1}^{n} \frac{(x_i - \bar{x}) \cdot (y_i - \bar{y})}{n-1}$

Population Covariance $\sigma_{xy} = \sum_{i=1}^{N} \frac{(x_i - \mu_x) \cdot (y_i - \mu_y)}{N}$

<2> Correlation :

Correlation is a measure of the Joint Variability of two Variables.

Unlike Covarince, Correlation could be thought of as a Standardized Measure.

It takes on values between -1 and 1, thus it is easy to Interpret the result.

↳ A Correlation of 1, known as Perfect Positive Correlation, means that one variable is perfectly explained by the other.

↳ A Correlation of 0 means that the Variables are Independent.

↳ A Correlation of -1, known as Perfect Negative Correlation, means that one Variable is explaining the other one perfectly, but they move in Opposite Directions.

Sample Correlation $r = \dfrac{S_{xy}}{S_x \cdot S_y}$

Population Correlation $\rho = \dfrac{\sigma_{xy}}{\sigma_x \cdot \sigma_y}$